**6** The isotope Iron-59 is a β-emitter with a half-life of 45 days. In order to estimate engine wear, an engine component is manufactured from iron throughout which the isotope Iron-59 has been uniformly distributed. The mass of the engine component is 2.4 kg and its initial activity is $8.5 \times 10^{7}$ Bq.
The component is installed in the engine 60 days after manufacture of the component, and then the engine is tested for 30 days. During the testing period, any metal worn off the component is retained in the surrounding oil. Immediately after the test, the oil is found to have a total activity of 880 Bq.
Calculate

**(a)** **(i)** the decay constant for the isotope Iron-59,

Decay constant $\lambda = \dfrac{\ln 2}{t_{1/2}}$

$= 0.0154 \text{ day}^{-1}$
$= 1.78 \times 10^{-7} \text{ s}^{-1}$

decay constant = ............................ $s^{-1}$ [2]

**(ii)** the total activity of the component when it was installed,

$A = A_o e^{-\lambda t} = 8.5 \times 10^{7} e^{-0.0154(60)}$
$A = 3.37 \times 10^{7}$ Bq

activity = ............................ Bq [2]

**(iii)** the mass of iron worn off the component during the test.

Activity after 90 days,

$A = 8.5 \times 10^{7} e^{-0.0154(90)}$
$= 2.1256 \times 10^{7}$ Bq

Mass of iron worn off $= \dfrac{880}{2.1256 \times 10^{7}} \times 2.4$ kg

$= 9.94 \times 10^{-5}$ kg

mass of iron = ............................ g [4]

**(b)** State and explain how your results in **(a) (iii)** will change if background radiation is included in the total activity of 880 Bq.

The mass of iron worn off the component in (a) (iii) will be less.

As background radiation is included in the total activity of 880 Bq, the actual activity due to Iron-59 in the oil is less than 880 Bq which means lesser mass of iron is worn off the component.

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## Section B

Answer **one** question from this Section in the spaces provided.

**7** **(a)** A pair of identical speakers, $S_1$ and $S_2$, 1.2 m apart makes up a stereo system in